

Feature Engineering and Encoding Report

Mobile Price Prediction Project | Task TA-007

1. Business Requirement

The mobile-phone company needs a machine-learning solution that classifies phones into price categories using technical specifications such as RAM, storage, battery capacity, camera quality, screen resolution, processor characteristics, and connectivity.

2. Task Objective

Task TA-007 converts the cleaned specification data into meaningful model features, creates additional summary features, encodes the classification target, and produces a dataset ready for train-test splitting and Random Forest classification.

3. Dataset Used

Input file: cleaned_dataset.csv. The file contains 1,017 records and the target column Price_Category. The source price is deliberately excluded from model inputs because it is used to create the target category.

4. Outcome

The final output contains 26 encoded model features and the encoded target column Price_Category_Encoded. Missing values after preparation: 0.

Engineered Features

Mobile Price Prediction Project | Task TA-007

Feature	Purpose	Business Meaning
Resolution_Megapixels	Width x height / 1,000,000	Overall screen detail
Total_Camera_MP	Rear camera + front camera	Combined camera capability
Battery_per_Screen_Inch	Battery / screen size	Battery capacity relative to display
Storage_per_RAM_GB	Storage / RAM	Storage capacity relative to memory
Resolution_Aspect_Ratio	Width / height	Display shape and viewing format

The engineered and encoded features are saved in ml_ready_dataset.csv.

Encoding and Feature Selection

Mobile Price Prediction Project | Task TA-007

Area	Decision	Reason
Numeric features	Median imputation	Keeps numeric model inputs complete and stable
Categorical feature	One-hot encoding with unknown category	Converts categories to model-readable binary format
Target encoding	0, 1, 2 labels	Provides stable class labels for classification
Excluded columns	Name, img, price	Avoids identifiers, URLs, and target leakage
Connectivity	Existing binary flags retained	Represents 3G, 4G, 5G, VoLTE, Wi-Fi, and Bluetooth

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Target Encoding and Quality Results

Mobile Price Prediction Project | Task TA-007

Price category	Encoded value	Records
Below 15,000	0	288
15,000-29,999	1	368
30,000 and above	2	361
Total	-	1,017

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Implementation Summary

Mobile Price Prediction Project | Task TA-007

Python Deliverable

feature_engineering_encoding.py reads cleaned_dataset.csv, validates required columns and target values, engineers specification features, applies preprocessing, encodes the target, and writes ml_ready_dataset.csv.

Validation Performed

The pipeline was executed successfully. It processed 1,017 records, produced 26 model features, generated a 27-column output including the target, and reported zero missing values after encoding.

Expected Next Step

The generated ml_ready_dataset.csv can now be divided into training and testing sets for Task TA-008, followed by Random Forest training and evaluation.

Conclusion

Task TA-007 is complete. The technical specifications have been transformed into consistent, interpretable, and machine-learning-ready inputs aligned with the business requirement of mobile price-category prediction.